Joshua Aarons

Doctoral Candidate — Economics Department — University of California San Diego

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Education

Ph.D. Candidate in Economics Expected completion: June 2027
University of California San Diego

B.A. in Economics 2016
Carleton College

Fields

Environmental Economics, Labor Economics, and Public Economics

Dissertation Committee

Judson Boomhower (Chair) Mark Jacobsen Joshua Graff Zivin Itzik Fadlon

Job Market Paper

"What's Driving High Electric Vehicle Fast Charging Prices?"

Electric vehicle fast charging is more costly than home charging, which could slow electric vehicle adoption and place higher costs on low-income households. I compile novel data on station quantities, prices, and marginal costs to shed light on price setting behavior across charging networks. Using a difference-in-differences approach, I show that contrary to standard theory, stations do not lower prices in response to nearby competitor entry. I build a model of heterogeneous consumer demand for fast charging and use the structurally-estimated parameters to calculate the social surplus maximizing prices. I show that charging prices include a markup above what is necessary to recover fixed costs, and estimate that lowering prices to efficient prices could increase social surplus by 8-10% while still recovering fixed costs. Efficient prices remain above marginal operating costs due to capacity constraints, which require stations to raise prices to clear the market. My results suggest that increasing competition can reduce markups by up to half, but that policy would need to address charging supply or inelastic consumer demand to further reduce markups.

Working Papers

- "Takeoffs and Test Scores: The Impact of Noise Pollution on Standardized Test Performance"

Noise pollution is ubiquitous in urban life, but little is known about its effects on educational attainment. I estimate the causal effect of noise pollution on standardized test performance by isolating exogenous variation in aircraft noise pollution and implementing a value-added model to account for differences in student ability. I find a large negative impact on math test performance, but no impact on English Language Arts test performance. Moving from a school at the 25th percentile of exposure to one at the 75th percentile is associated with a math test score reduction of 0.1 standard deviations, twice as large as the impact of missing 10 school days during the year. I find similar results using a second identification strategy that leverages changes in aircraft noise from the introduction of new flight paths. These findings suggest a potentially large value to investments in noise insulation for schools exposed to high levels of noise pollution.

Relevant Positions Held

Graduate Student Researcher, Judson Boomhower 2022-2025

- Compiled insurance rates and hurricane risk by scraping insurance and categorical risk model filings.
- Conducted analyses of insurer's pricing of wildfire and hurricane risk.
- Aggregated and cleaned oil and gas well records to build panels of well ownership and production.

Instructor, UC San Diego 2025

- Served as instructor for ECON 5: Introduction to Social Data Analytics.
- Planned and led classes and developed assignments and other course materials.

Teaching Assistant, UC San Diego 2022-present

- Planned and led discussion sessions and worked one-on-one with students for classes on environmental economics, intermediate microeconomics, and data analysis.
- Developed a curriculum of academic readings for a class on environmental economics.

Health Policy Center, Urban Institute 2017-2021
Research Analyst 2019-2021
Research Assistant 2017-2019

- Used Medicare administrative data to compare home health and post-acute care use patterns across different Medicare insurance plan characteristics.
- Assessed determinants of access to mental health and substance use disorder care.
- Derived propensity score and entropy balance weights for an evaluation of care models for Medicare and Medicaid beneficiaries.
- Provided descriptive analysis of differences across populations in outcomes such as access to health insurance, mental health care, and exposure to COVID-19.

Fellowships, Honors, and Awards

Benjamin C. Horne Memorial Prize, UC San Diego Department of Economics 2025
Clive Granger Research Fellowship, UC San Diego Department of Economics 2025
Summer Graduate Teaching Scholar – Outstanding Scholar Award, UC San Diego 2025
Teaching Assistant Excellence Award, UC San Diego Department of Economics 2025

Presentations

Economics of EV Charging Workshop, The Salata Institute for Climate and Sustainability 2026
Annual Conference, Western Economic Association International 2026
SustainableED Conference, Annenberg Institute at Brown University 2026
AERE Summer Conference, Association of Environmental and Resource Economists 2024

Other Information

Citizenship: United States
Skills: R, Python, Stata